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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (E&E) (Credit System) Degree Examinations
 Supplementary Examinations - December 2025

21EE8X79 / 20EE8X79 - ELECTRIC VEHICLE TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain Newton's second law of motion applied to the vehicle.	6	L*2	1	1, 2
	b) With neat diagram explain the configuration of electric vehicle.	7	L2	2	1, 2, 3
	c) Explain the parameters on which the vehicle performance can be calculated.	7	L2	2	1, 2, 3
2.	a) Explain how to calculate the torque at the wheels and vehicle speed.	7	L2	1	1, 2
	b) Explain the roadway maximum percent grade with equation.	7	L3	1	1, 2
	c) Explain how to calculate tractive effort and vehicle speed.	6	L2	2	1, 2, 3
3.	a) Explain parallel hybrid electric vehicle with diagram.	8	L2	2	1, 2, 3
	b) Explain the plot of a velocity profile for varying level road case.	6	L3	1	1, 2
	c) Explain the typical electric motor torque-speed characteristics.	6	L2	2	1, 2, 3
Unit – II					
4.	a) With neat diagram explain the components of a battery cell.	6	L2	3	1, 2, 3
	b) With neat graph explain torque speed characteristics of SRM drive.	7	L2	4	1, 2, 3
	c) Explain the methods to control the speed of DC motor drives.	7	L2	4	1, 2, 3
5.	a) Explain how to calculate theoretical capacity of battery in Ah.	7	L2	3	1, 2, 3
	b) Explain the types of rechargeable batteries for EV and HEV.	7	L2	3	1, 2, 3
	c) With neat diagram explain power electronics-controlled DC motor drive.	6	L2	4	1, 2, 3
6.	a) What are the requirements of an EV and HEV motor.	6	L2	4	1, 2, 3
	b) Explain in brief the different types of fuel cells.	8	L3	3	1, 2, 3
	c) With neat diagram explain the permanent magnet motor.	6	L2	4	1, 2, 3
Unit – III					
7.	a) With neat diagram explain thermostat control strategy.	7	L3	5	1, 2, 3
	b) Explain the optimal operating region of the hybrid electric vehicle.	6	L3	5	1, 2, 3
	c) Explain how to calculate the power capacity of PPS to fully utilize the electric motor capacity.	7	L2	5	1, 2, 3
8.	a) Explain Max-SOC-of-PPS control strategy.	7	L2	5	1, 2, 3
	b) With neat diagram explain the configuration of a typical series hybrid electric drive train:	7	L2	5	1, 2, 3
	c) With neat diagram explain the tractive effort vs. vehicle speed characteristics of an electric motor-driven vehicle.	6	L2	5	1, 2, 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
